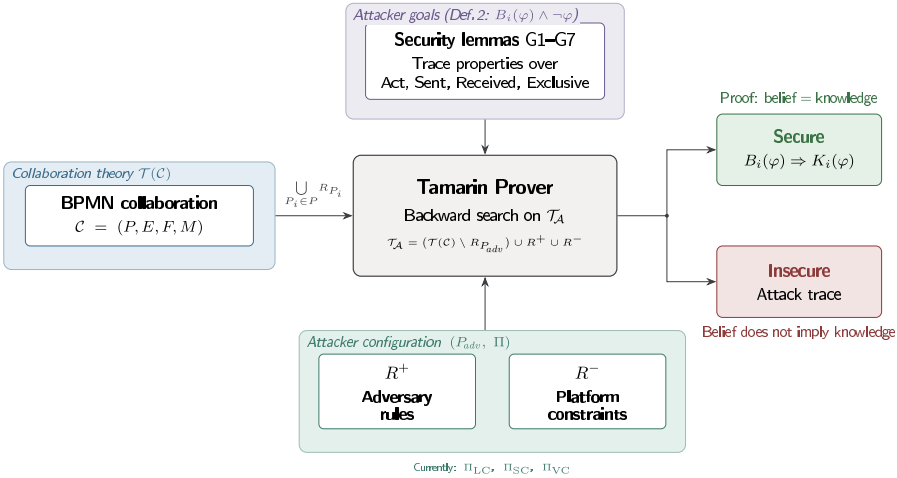
Attacker goals (Def. 2: $B_i(\varphi) \wedge \neg\varphi$)
Security lemmas G1–G7
Trace properties over
Act, Sent, Received, Exclusive
Collaboration theory $\mathcal{T}(\mathcal{C})$
BPMN collaboration
$\mathcal{C} = (P, E, F, M)$
$\bigcup_{P_i \in P} R_{P_i}$
Tamarin Prover
Backward search on $\mathcal{T}_\mathcal{A}$
$\mathcal{T}_\mathcal{A} = (\mathcal{T}(\mathcal{C}) \setminus R_{P_{adv}}) \cup R^+ \cup R^-$
Proof: belief = knowledge
Secure
$B_i(\varphi) \Rightarrow K_i(\varphi)$
Insecure
Attack trace
Belief does not imply knowledge
Attacker configuration (P_{adv}, Π)
R^+
Adversary rules
R^-
Platform constraints
Currently: Π_{LC}, Π_{SC}, Π_{VC}